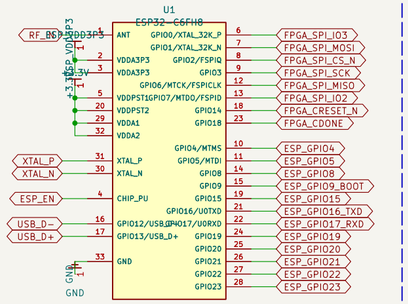
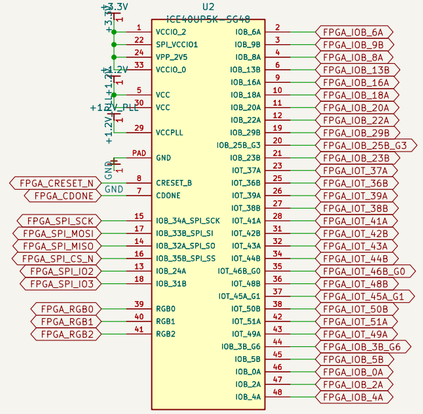
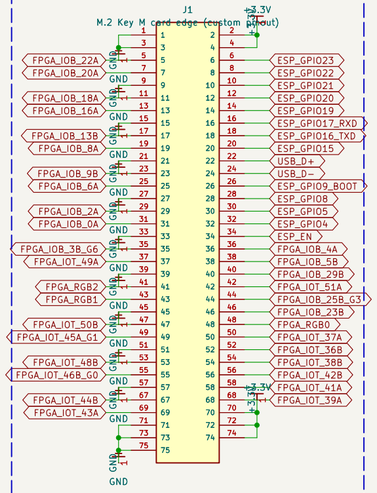
ESP32-C6FH8 (QFN32, 8 MB in-package flash)



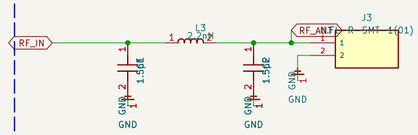
iCE40UP5K-SG48 (3.3 V I/O, 1.2 V core)



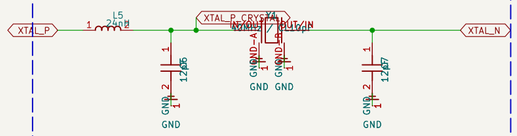
M.2 key M card edge J1 (custom pinout, 3.3 V in, see docs/icd-v2-pinout.csv)



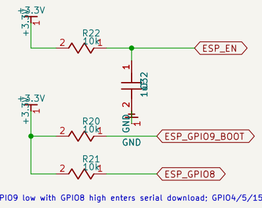
RF front end (initial C-L-C match, tune on hardware)



40 MHz crystal (24 nH series on XTAL_P per C6 checklist)

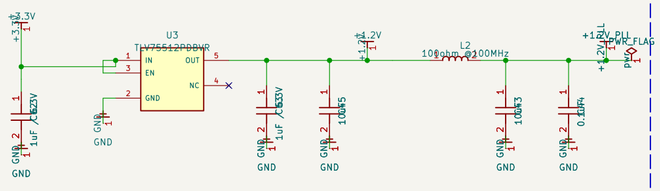


Reset and boot straps

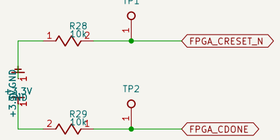


GPIO9 low with GPIO8 high enters serial download; GPIO4/5/15 straps left at chip defaults.

1.2 V FPGA core regulator

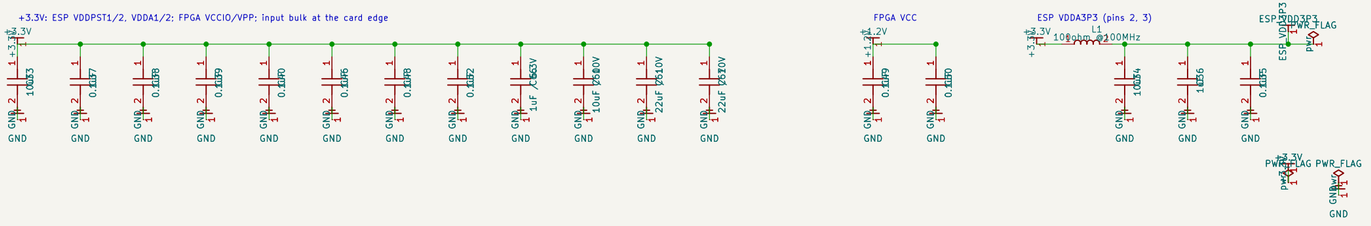


FPGA configuration link



RGB0-2 (FPGA current-sink drivers, not plain GPIO) go straight to contacts 48/43/41; the v1 on-module LED is gone.

Decoupling and supply filters



PWR_FLAGs: +3.3V and GND enter on card-edge contacts 2/4/70/72/74 and 1/3/9/15/21/27/33/39/45/51/57/71/73/75.

Sheet: /		
File: module.kicad_sch		
Title: ESP32-C6 + iCE40UP5K M.2 module v2		
Size: User	Date:	**Rev: B — NOT FOR FABRICATIO**
KiCad E.D.A. 10.0.6		Id: 1/1